



DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AIML,AIDS, CS(AIML),CS(CY)31	Sub:	Laplace Transforms and Vector Spaces	Test:	01
Time:	2.00 -3.00 pm	Term:	03.10.2024 to 25.01.2025	Marks:	30
Date:	28-11-2024	Semester:	IIII	Section:	CS/IS/AIML/AIDS, CS(AIML),CS(CY)

Note: Answer any TWO full questions. Each main question carries 15 marks

Q. No.	Questions	Bloom Level	CO's	Marks
1	a) Find $L\left\{\frac{\cos 2t - \cos 3t}{t} + t \sin t\right\}$	L2	CO1	5
	b) Express the function $f(t) = \begin{cases} t^2, & 0 < t < 2, \\ t-1, & 2 < t < 4, \\ 7, & t > 4. \end{cases}$ in terms of the Heaviside function and hence find its Laplace transform.	L3	CO2	5
	c) Find $L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$ using convolution theorem.	L4	CO2	5
2	a) Find (i) $L\left\{\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^2\right\}$ (ii) $L\left\{\int_0^t e^{-2t} \sin 3t \, dt\right\}$.	L1	CO1	5
	b) Find $L\{f(t)\}$ of a periodic function of period $2a$ is defined by $f(t) = \begin{cases} E & \text{for } 0 \leq t \leq a \\ -E & \text{for } a \leq t \leq 2a \end{cases}$.	L3	CO2	5
	c) Solve $y'' + 4y' + 3y = e^{-t}$ given that $y(0) = 1, y'(0) = 1$ by the method of Laplace transform.	L4	CO2	5
3	a) If $L\{f(t)\} = F(s)$, then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, where n is a positive integer.	L1+L2	CO1	5
	b) Find of (i) $L^{-1}\left\{\frac{se^{-as} + 1}{s^2 + \omega^2}\right\}$ (ii) $L^{-1}\left\{\log\left(\frac{s+4}{s-4}\right)\right\}$.	L3	CO2	5
	c) Solve the simultaneous equations $\frac{dx}{dt} - 2y = \cos 2t; \frac{dy}{dt} + 2x = \sin 2t$, given $x(0) = 1, y(0) = 0$ using Laplace transform.	L5	CO2	5